Experiment 6:
Develop Pig Latin scripts to sort, group, join, project, and filter the data.

Assumed Weather Data (weather_data.txt):

weather_data.txt

```
2025-03-31,38,70,Hot Day
2025-03-30,8,60,Cold Day
2025-03-29,22,85,Humid Day
2025-03-28,25,50,Normal Day
2025-03-27,40,90,Hot Day
2025-03-26,5,30,Cold Day
```

Upload weather_data.txt file to hdfs.
Create a directory and upload weather_data.txt file to hdfs folder.

```
[training@localhost ~]$ hadoop dfs -mkdir /pig
[training@localhost ~]$ hadoop dfs -put /home/training/Desktop/weather_data.txt
/pig/
```

Pig Latin Script in CommandLine:
Loading the Data:
weather = LOAD '/pig/weather_data.txt' USING PigStorage(',') AS (date:CHARARRAY, temperature:INT, humidity:INT, condition:CHARARRAY);

1. Sorting Data (By Temperature)
sorted_weather = ORDER weather BY temperature DESC;

DUMP sorted_weather;

Output:

```
(2025-03-27,40,90,Hot Day)
(2025-03-31,38,70,Hot Day)
(2025-03-28,25,50,Normal Day)
(2025-03-29,22,85,Humid Day)
(2025-03-30,8,60,Cold Day)
(2025-03-26,5,30,Cold Day)
```

2. Grouping Data (By Weather Condition)
grouped_weather = GROUP weather BY condition;

DUMP grouped_weather;

Output:

```
(Hot Day,{(2025-03-31,38,70,Hot Day),(2025-03-27,40,90,Hot Day)})
(Cold Day,{(2025-03-30,8,60,Cold Day),(2025-03-26,5,30,Cold Day)})
(Humid Day,{(2025-03-29,22,85,Humid Day)})
(Normal Day,{(2025-03-28,25,50,Normal Day)})
```

3. Joining Data (Example: Joining Weather with Another Dataset)
Let's assume we have another dataset **weather_info.txt**:

weather_info - Notepad
File Edit Format View Help

```
Hot Day,Wear Light Clothes
Cold Day,Wear Warm Clothes
Humid Day,Stay Hydrated
Normal Day,Enjoy the Weather
```

upload weather_info.txt file to hdfs folder.

```
[training@localhost ~]$ hadoop dfs -put /home/t
raining/Desktop/weather_info.txt /pig/
[training@localhost ~]$
```

weather_info = LOAD '/pig/weather_info.txt' USING PigStorage(‘,') AS (condition:CHARARRAY, advice:CHARARRAY);

joined_data = JOIN weather BY condition, weather_info BY condition;

DUMP joined_data;

Output:

```
2025-03-31,38,70,Hot Day,Hot Day,Wear Light Clothes)
2025-03-27,40,90,Hot Day,Hot Day,Wear Light Clothes)
2025-03-30,8,60,Cold Day,Cold Day,Wear Warm Clothes)
2025-03-26,5,30,Cold Day,Cold Day,Wear Warm Clothes)
2025-03-29,22,85,Humid Day,Humid Day,Stay Hydrated)
2025-03-28,25,50,Normal Day,Normal Day,Enjoy the Weather)
```

4. Projection (Selecting Specific Columns)
projected_weather = FOREACH weather GENERATE date, temperature;

DUMP projected_weather;
Output (Only Date & Temperature):

```
(2025-03-31,38)
(2025-03-30,8)
(2025-03-29,22)
(2025-03-28,25)
(2025-03-27,40)
(2025-03-26,5)
```

5. Filtering Data (E.g., Find Hot Days)
hot_days = FILTER weather BY temperature > 35;

DUMP hot_days;
Output (Only Hot Days, Temp > 35):

```
(2025-03-31,38,70,Hot Day)
(2025-03-27,40,90,Hot Day)
```

Function	Pig Script
Sort	ORDER weather BY temperature DESC;
Group	GROUP weather BY condition;
Join	JOIN weather BY condition, weather_info BY condition;
Project	FOREACH weather GENERATE date, temperature;
Filter	FILTER weather BY temperature > 35;